A PROJECT

ON

Financial Sentiment Analysis Using NLP and BERT for Market Prediction

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

AmruthaVarshini J **2520080066**

Jhanasri Thota **2520090121**

Under the Esteemed guidance of

Dr.Y. Lakshmi Prasanna

Assistant Professor, CSIT

Department Of CS-IT & AI-DS



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

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DEPARTMENT OF AI-DS AND CS-IT

Name of Title: Financial Sentiment Analysis Using NLP and BERT for Market Prediction

1. ABSTRACT:

The financial market is influenced by several factors, including company announcements, economic reports, global events, and financial news. Every day, thousands of news articles, blogs, and market updates are published, making it difficult for investors and financial analysts to manually read and analyze all the available information. The sentiment expressed in financial news often affects investor confidence and decision-making, which in turn can influence stock market trends. Therefore, automatically analyzing financial text has become an important application of Artificial Intelligence and Natural Language Processing (NLP).

The main objective of this project is to develop a **Financial Sentiment Analysis** system using **Natural Language Processing (NLP)** and the **BERT (Bidirectional Encoder Representations from Transformers)** model. The proposed system will classify financial news into three sentiment categories: positive, negative, and neutral. This classification will help in understanding the overall sentiment of financial news and its possible impact on market behaviour. To evaluate the effectiveness of different approaches, traditional machine learning models will first be implemented as baseline models before applying the BERT model. The performance of these models will be compared using evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrix**.**

The project will use a publicly available financial news dataset for training and testing purposes. The overall process includes data collection, text preprocessing, feature extraction, model training, performance evaluation, and result analysis. The project will also study how transformer-based language models perform compared to conventional machine learning techniques in sentiment classification tasks.

This project provides practical exposure to machine learning, deep learning, and modern NLP techniques while helping students understand how Artificial Intelligence can process and analyze textual data efficiently. The developed system can support investors, financial analysts, and researchers by providing quick insights into market sentiment, reducing the time required for manual analysis, and assisting in making more informed financial decisions. Additionally, the knowledge gained through this project can serve as a foundation for future research in financial analytics, intelligent decision support systems, and AI-driven market prediction.

2. INTRODUCTION:

Every day, financial markets generate a huge amount of information through news articles, company announcements, reports, and social media. The sentiment expressed in this information often influences investor decisions and market behaviour. However, manually reading and analysing such a large volume of financial text is not practical. Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to understand and process human language. Recent advancements in NLP, especially the BERT model, have greatly improved the ability of computers to understand the context and meaning of text.

The main objective of this project is to build a system that can automatically classify financial news as positive, negative, or neutral. The project will compare traditional machine learning techniques with the BERT model to study their performance and effectiveness. This comparison will help identify which approach is better suited for financial sentiment analysis while also providing a better understanding of modern NLP techniques.

3. SOFTWARE REQUIREMENTS:

Category	Details
Operating System	Windows 11
Programming Language	Python
Development Tool	Visual Studio Code
Libraries	Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, NLTK, Transformers (Hugging Face), PyTorch
Dataset	Financial PhraseBank Dataset (or another publicly available financial sentiment dataset)
Version Control	Git and GitHub

Signature of the Guide **Course Instructor**